

```
theory Week_1_sol
imports Main
begin
```

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Term 1

Computer-Assisted Theorem Proving: 7CCSACTP

Exercise Sheet 1

Important Note: Please download Isabelle from the link provided in the slides and bring a laptop to the large group tutorial. Performing the proofs in Isabelle will be an integral part of those tutorials.

Before you start, please open a new Isabelle/HOL theory file called Week_1.thy and add the following to its beginning.

```
theory Week_1
imports Main
begin
```

Exercise 1.1 Trying out Isabelle

In this sheet, you should try out the Isabelle theorem prover. Prove the following theorem, first on pen-and-paper, then formally in Isabelle.

```
lemma
  assumes T: T b
    and A: A a  $\wedge$  A b  $\implies$  a = b
    and TA:  $\bigwedge x. T x \implies A x$ 
    and Aab: A a
  shows T a
```

proof—

```
  have 1: A b
    using TA[OF T]
    .
```

```
  have 2: A a  $\wedge$  A b
```

```

    apply(rule conjI)
    using Aab 1
  .

  have 3:  $a = b$ 
    apply(rule A)
    using 2
  .

  show ?thesis
    using T
    apply(subst 3)
  .
qed

end

```